

Binomial Distribution

2 possible outcomes: True or False, Success or Failure, Yes or No

Every trial is independent trial which means the outcome of one trial does not affect the outcome of another trial

Binomial $\binom{n}{x} = \frac{n!}{x!(n-x)!}$ # of experiments

$P \Rightarrow$ Probability of success

$Q \Rightarrow$ Probability of failure $\Rightarrow 1-p$

✓ Mean $\Rightarrow \mu = np$

✓ Variance $\Rightarrow \sigma^2 = npq$

SD $\Rightarrow \sigma = \sqrt{npq}$

Given Consider 3 Bernoulli trials for tossing a coin. H $\rightarrow$ Success T $\rightarrow$ Failure

3 ways in which we can have 2 success in 3 trials {SFF, FSF, FFS}

1 No. of successes for the above cases can take 4 values 0, 1, 2, 3

a $\rightarrow$ Prob of success b $\rightarrow$ Prob of failure

$$P(X=0) = P(FFF) = P(F) \times P(F) \times P(F) \\ \Rightarrow b^3$$

$$P(FFF) = P(F) \times P(F) \times P(F) \\ = b^3$$

$$P(X=1) = P(SFF, FSF, FFS) = P(S) \times P(F) \times P(F) + P(F) \times P(S) \times P(F) + P(F) \times P(F) \times P(S) \\ = 3ab^2$$

$$P(X=2) = 3a^2b$$

$$P(X=3) = a^3$$

$$(a+b)^3 \Rightarrow a^3 + 3ab^2 + 3a^2b + b^3$$

$$(a+b)^3$$

$$P(X=3) = a^3$$

$$P(X=0) = b^3 \quad P(X=1) = 3ab^2 \quad P(X=2) = 3a^2b$$

Coin tossed 8 times $P(X=5)$
 $\rightarrow$ Exactly 5 heads $\Rightarrow {}^8C_5 a^5 b^{8-5}$

$$P(a) = \frac{1}{2} \quad P(b) = \frac{1}{2} \Rightarrow \frac{8!}{3!5!} \times \left(\frac{1}{2}\right)^5 \times \left(\frac{1}{2}\right)^3$$

$$= \frac{8!}{3!5!} \times \left(\frac{1}{2}\right)^8 = \frac{7}{32} = \frac{219}{256}$$

Poisson Distribution

→ It's concerned with discrete random variables

→ Describe # of events occurring in fixed time interval

of people visiting hospital in every hour

$$P(x, \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$e \Rightarrow$ base of logarithm

$x \Rightarrow$ # of occurrences

$\lambda \Rightarrow$ Expected # of event occur at every

Parameters

$\lambda \in (0, \infty)$ (rate)

Support

$k \in \mathbb{N}$ (Natural numbers starting from 0)

PMF

$$\frac{\lambda^k e^{-\lambda}}{k!}$$

CDF

$$\frac{\Gamma(Lk+1, \lambda)}{Lk!}$$

$$e^{-\lambda} \sum_{j=0}^{Lk} \frac{\lambda^j}{j!}$$

$$Q(Lk+1, \lambda)$$

Mean

$$\lambda$$

Median

$$\left[\lambda + \frac{1}{3} - \frac{1}{50\lambda} \right]$$

Mode

$$\lceil \lambda - 1 \rceil, \lfloor \lambda \rfloor$$

Variance

$$\lambda$$

As only 3 students came to attend the class today, find the probability for exactly 4 students to attend the classes tomorrow.

Avg. rate of value $\Rightarrow 3$

$$\lambda = 3$$

Poisson Random Variable $\Rightarrow$

$$X = 4$$

$$P(X=4) = \frac{e^{-3} \cdot 3^4}{4!} = 0.168$$

Without replacement

After 1 white draw:-

9 white + 10 black left (19)

Now Black removed

10 white + 9 black

$$P(W_2|W_1) = \frac{9}{19} \approx 0.4737$$

$$P(W_2|B_1) = \frac{10}{19} \approx 0.5263$$

Probability of success is not constant

from bag containing 10 white and 10 black balls. Predict whether the draw is replaced and not replaced.

Ans:

probability of success :- $p = \frac{10}{20} = \frac{1}{2}$

placement:

$$P(\text{Success}) \text{ for trial 1} = \frac{1}{2}$$

$$\text{for trial 2} = \frac{9}{19}$$

Not Bernoulli trial

Gaussian Distribution

$$N(\mu, \sigma^2)$$

Parameters

$\mu \in \mathbb{R} = \text{mean}$

$\sigma^2 \in \mathbb{R}_{>0} = \text{variance}$

Support

$x \in \mathbb{R}$

PDF

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Error function

CDF

$$\Phi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{2} \left[1 + \text{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right) \right]$$

Quantile

$$\mu + \sigma\sqrt{2} \text{erf}^{-1}(2p-1)$$

Mean, Median, Mode

μ

Variance

$$\sigma^2 = \text{Var}(X)$$

Calculate PDF of Gaussian distrib
using the following data

$$x=2, \mu=5 \text{ and } \sigma=3$$

$$e^{-\frac{(x-\mu)^2}{2\sigma^2}} \Rightarrow e^{-\frac{(2-5)^2}{18}}$$

$$\Rightarrow \frac{1}{(3\sqrt{2\pi})} \times 0.6065$$

$$\Rightarrow 0.1328 \times 0.6065$$

$$\Rightarrow 0.0805$$

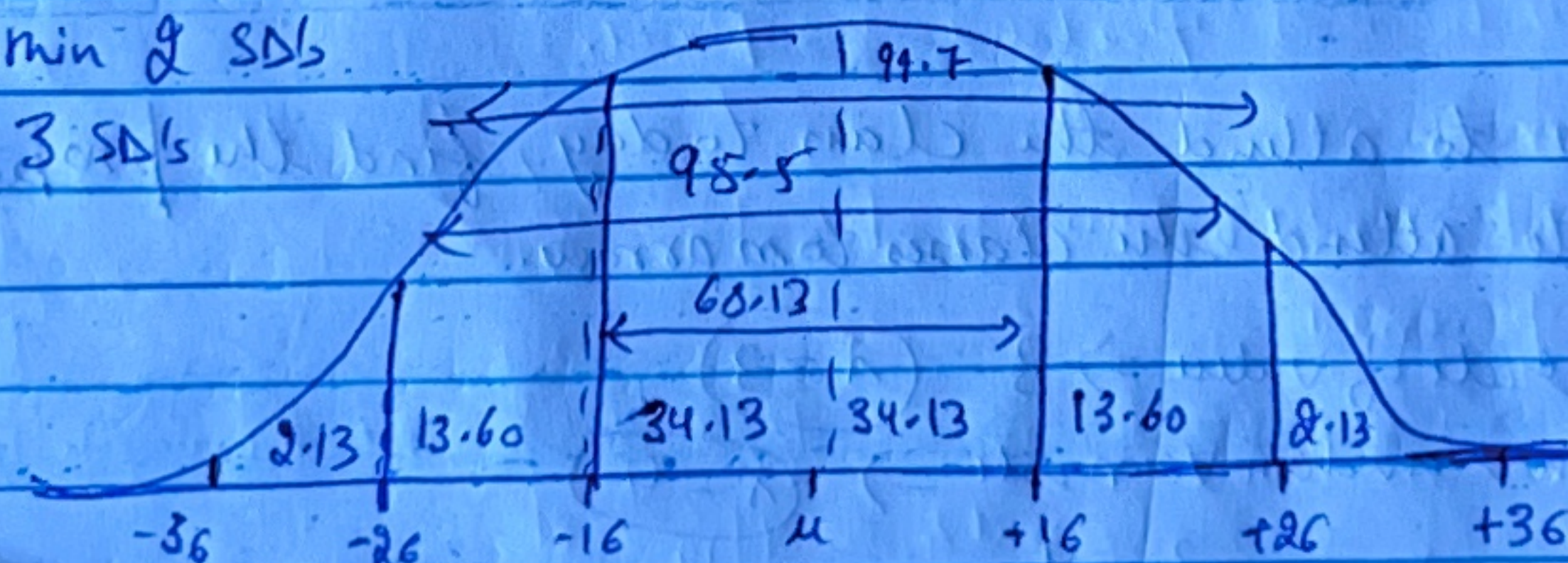
Empirical Rule of Normal Distribution

Also known as 68-95-99 rule

68% $\Rightarrow$ Observed data points will lie inside one SD of the mean

95% $\Rightarrow$ fall within 2 SDs

99.7% $\Rightarrow$ within 3 SDs



68.13 $\rightarrow$ 1st SD of 1st mean

95.45 $\Rightarrow$ 2nd SD of 2nd mean

99.73 $\Rightarrow$ 3rd SD of 3rd mean